Question

Solid $\mathbf{X}$ is a crystal with a ZnS structure. $\mathbf{X}$ can be synthesized through aqueous solution reactions: Oxide $\mathbf{A}$ (containing 25.6% oxygen) is mixed with element $\mathbf{B}$, which has a tetrahedral molecular structure, in dilute NaOH aqueous solution and heated to react. $\mathbf{A}$ dissolves to obtain $\mathbf{C}$ (Reaction 1), and $\mathbf{B}$ disproportionates to obtain $\mathbf{D}$ (the conversion rate of the reaction is 25% (Reaction 2); then $\mathbf{C}$ reacts with $\mathbf{D}$ to obtain $\mathbf{X}$ (Reaction 3). Adding an appropriate amount of iodine to the system can increase the reaction rate and the yield of $\mathbf{X}$, possibly because: $\mathbf{B}$ and elemental iodine undergo a new reaction in dilute NaOH aqueous solution in a molar ratio of 1:2, generating products such as $\mathbf{D}$ and $\mathbf{E}$ (the molar ratio of the products is 1:1) (Reaction 4). In addition, $\mathbf{X}$ can also be synthesized by traditional methods at high temperatures. Reacting $\mathbf{A}$ with $\mathbf{D}$ at $950^{\circ}\mathrm{C}$ under a $\mathrm{H_2}$ atmosphere (Reaction 5), or reacting $\mathbf{B}$ with the low-valent chloride $\mathbf{F}$ at $1050^{\circ}\mathrm{C}$ in vacuum (Reaction 6), can both obtain $\mathbf{X}$. $\mathbf{F}$ is a gaseous molecule at high temperature, and it rapidly cools to 77K and transforms into a red solid; upon further heating to above $0^{\circ}\mathrm{C}$, $\mathbf{F}$ undergoes disproportionation decomposition.

The following statements are made:

1, Let the atomic numbers of the elements in $\mathbf{X}$ be a and b (a>b), then $\frac{a-b}{a+b} = 0.35$.

2, The mass fractions of hydrogen in $\mathbf{C}$ and $\mathbf{D}$ are 2.508% and 8.896%, respectively.

3, The purpose of evacuating in Reaction 6 is to reduce the boiling point of the reactants.

4, Among Reactions 1-6, Reaction 6 has the smallest difference in the number of molecules between reactants and products.

Then, which of the following options contains all the correct statements (a calculation result error within 0.01 in statement 1 is considered correct):

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1, 2

G. 1, 3

H. 1, 4

I. 2, 3

J. 2, 4

K. 3, 4

L. 1, 2, 3

M. 1, 2, 4

N. 1, 3, 4

O. 2, 3, 4

P. 1, 2, 3, 4

Answer

Correct Answer: $\mathbf{M}$

Detailed Explanation

Starting from oxide $\mathbf{A}$, let the chemical formula of $\mathbf{A}$ be $\mathbf{M}_2\mathrm{O}_y$, therefore:

$$\frac{16y}{2M+16y} = 25.6\%$$

Starting from $y = 1$, when $y = 3$, we solve for $M = 69.75\,g/mol$. Considering the precision, it can be considered as the element Ga. Therefore, $\mathbf{A}$ is $\mathrm{Ga_2O_3}$.

CHECKPOINT 1 PTS

$\mathbf{A}$ is $\mathrm{Ga_2O_3}$

$\mathrm{Ga_2O_3}$ dissolves in sodium hydroxide solution to form $\mathrm{NaGa(OH)_4}$, so $\mathbf{C}$ is $\mathrm{NaGa(OH)_4}$.

CHECKPOINT 1 PTS

$\mathbf{C}$ is $\mathrm{NaGa(OH)_4}$

The tetrahedral molecular element $\mathbf{B}$ is easily obtained as $\mathrm{P_4}$.

CHECKPOINT 0.5 PTS

$\mathbf{B}$ is P_4

The disproportionation reaction of P_4 under alkaline conditions is as follows:

$$P_4 + 3NaOH + 3H_2O \rightarrow PH_3 + 3NaH_2PO_2$$

It can be seen from the equation that the substance with a yield of 25% is PH_3, so $\mathbf{D}$ is PH_3.

CHECKPOINT 1 PTS

$\mathbf{D}$ is PH_3

According to the hints given in the question, reaction 4 can be written (the weakly alkaline product should be NaH_2PO_4):

$$P_4 + 2I_2 + 2H_2O + 6NaOH \rightarrow 2PH_3 + 2NaH_2PO_4 + 4NaI$$

Therefore, $\mathbf{E}$ is NaH_2PO_4.

CHECKPOINT 1 PTS

$\mathbf{E}$ is NaH_2PO_4

According to the conservation of elements, it can be inferred that $\mathbf{F}$ contains the element Ga, so it is a low-valent chloride of Ga, namely $GaCl$. Therefore, $\mathbf{F}$ is $GaCl$.

CHECKPOINT 1 PTS

$\mathbf{F}$ is GaCl

The following is an analysis of the options:

1. The atomic numbers of the elements in $\mathbf{X}$ are 31 and 15, so $\frac{a-b}{a+b}=0.35$. Correct.

2. The mass fractions of hydrogen in $\mathbf{C}$ and $\mathbf{D}$ are 2.508% and 8.896%, respectively. Correct.

3. The reactants in reaction 6 have low boiling points, but GaCl is easily oxidized, so vacuuming is to protect the reactants. Incorrect.

CHECKPOINT 1 PTS

GaCl is easily oxidized, and vacuuming can protect the reactants

4. First write out reactions 1-6:

Reaction 1: $\mathrm{Ga_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaGa(OH)_4}$

CHECKPOINT 1 PTS

Reaction 1: $\mathrm{Ga_2O_3 + 2NaOH + 3H_2O \rightarrow 2NaGa(OH)_4}$

Reaction 2: $\mathrm{P_4 + 3NaOH + 3H_2O \rightarrow PH_3 + 3NaH_2PO_2}$

CHECKPOINT 1 PTS

Reaction 2: $P_4 + 3NaOH + 3H_2O \rightarrow PH_3 + 3NaH_2PO_2$

Reaction 3: $NaGa(OH)_4 + PH_3 \rightarrow GaP + NaOH + 3H_2O$

CHECKPOINT 1 PTS

Reaction 3: $NaGa(OH)_4 + PH_3 \rightarrow GaP + NaOH + 3H_2O$

Reaction 4: $P_4 + 2I_2 + 2H_2O + 6NaOH \rightarrow 2PH_3 + 2NaH_2PO_4 + 4NaI$

CHECKPOINT 1 PTS

Reaction 4: $P_4 + 2I_2 + 2H_2O + 6NaOH \rightarrow 2PH_3 + 2NaH_2PO_4 + 4NaI$

Reaction 5: $Ga_2O_3 + 2PH_3 \rightarrow 2GaP + 3H_2O$

CHECKPOINT 1 PTS

Reaction 5: $Ga_2O_3 + 2PH_3 \rightarrow 2GaP + 3H_2O$

Reaction 6: $6GaCl + P_4 \rightarrow 4GaP + 2GaCl_3$

CHECKPOINT 1 PTS

Reaction 6: $6\mathrm{GaCl} + \mathrm{P_4} \rightarrow 4\mathrm{GaP} + 2\mathrm{GaCl_3}$

It can be seen immediately that the reaction with the smallest difference in the number of molecules between reactants and products is reaction 6. Correct.